Modular and Compositional Adaptation of a Multilingual LM

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August 16, 2026

1 Preliminaries

The core task is to adapt pretrained multilingual LMs independently for two factors, training a dedicated module for each, then to compose the two independently trained modules on the small intersection corpus and measure how well the resulting LM models that domain. Two choices run through every section below and are settled here: where the text comes from, and what we measure.

Why not the HuggingFace releases. Both datasets on HuggingFace are distributed as *sentence pairs*, one configuration per language pair, which is not very suitable for this task, for three reasons. First, what we need is a monolingual corpus per language, not the translation relations between languages; recovering one language means going through every configuration it appears in, taking its side of each, and then deduplicating, because the same sentence recurs across those pairs. Second, even once deduplicated the material is a set of individual sentences: too short for adapting or scoring a language model, which wants the longer coherent text it was pretrained on, and with no document to split along, so neighbouring sentences of one work can end up on both sides of a train/test boundary. Third, the coverage is partial: DGT is released as 10 pair configurations, against the 25 languages the corpus itself covers.

We therefore took both corpora as *raw documents* from the OPUS repository (Tiedemann, 2012), which publishes one XML file per document per language. That removes all three problems at once: each language is one self-contained set of documents, the text is document-length, and a split can be drawn between whole documents. Table 1 reports what we extracted, summed over all languages of each dataset. The two are very unlike each other, which is the point: 909k short legal documents against 158 books, and that difference is the genre factor this study tries to isolate.

Dataset	Languages	Documents	Sentences	Words
DGT	25	908,945	132.4M	2.07B
Books	16	158	0.9M	16.5M

Table 1: The statistics of the two datasets after extraction from the OPUS raw documents.

What we measure. The usual metric for a language model is perplexity, which is defined per token. That is not usable here, because the comparisons we care about run across models and across languages, and a token is not the same unit in any two of them: a tokenizer that cuts a language into many short pieces makes each one easier to predict and reports a lower perplexity for it, without the model being any better. We therefore report bits per byte (BPB) (Gao et al., 2020), which normalises the same likelihood over bytes instead of tokens. Bytes belong to the text itself rather than to whoever tokenized it, so no model gains from segmenting a language more finely than another does, and any two of them can be compared on the same footing. We use it both to choose the backbone and to score every model in this report.

2 Subtask 1: backbone and modular architecture selection

Backbone. We looked at three model families developed in different parts of the world, on the assumption that their pretraining mixtures differ structurally: EuroLLM (Martins et al., 2024) from Europe, built for all official European Union languages and so the one we expected to be strongest here; Qwen (Qwen Team, 2025) from Asia; and Llama (Grattafiori et al., 2024) from North America. We took two of each, one of 1.7 to 4B parameters and one of 8 to 9B, because a difference that only shows up at one scale is not a property of the family.

Rather than pick by reputation we measured

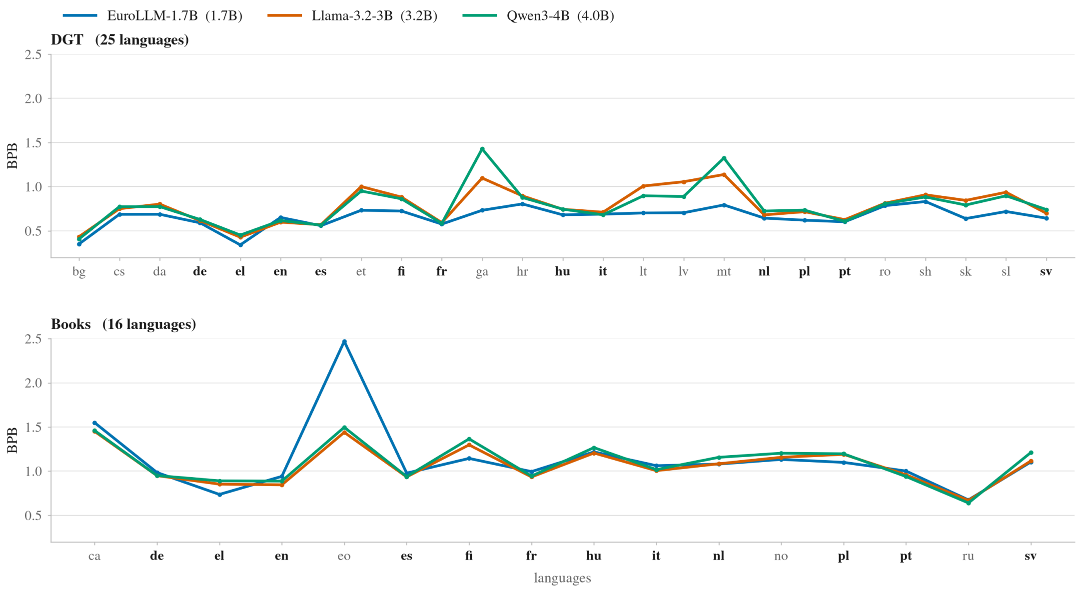

Figure 1: The three families before any adaptation, scored on every language of both datasets. The larger tier gives the same ordering.

what each model already knows about the two datasets. Each model, with no adaptation of any kind, was scored on every language of both datasets, 41 languages in all, each sampled down to about 250k words. All six models read exactly the same text, so what separates them is the model and nothing else. Figure 1 shows the smaller tier. The ordering is the same in both tiers and on both datasets: EuroLLM first, then Llama, then Qwen.

We ran the whole study on two backbones from the small tier, **EuroLLM-1.7B** and **Qwen3-4B**, to save time and compute budget. Within the tier we took the strongest and the weakest rather than two arbitrary models, assuming that a result which holds at both ends of the range we measured is unlikely to depend on how well the backbone already knew these datasets.

Modules and how they combine. These two choices are really one, because what makes an adapter easy to train separately is also what makes it easy to put back together. We use LoRA (Hu et al., 2021): a parameter-efficient modular architecture that leaves the backbone frozen and trains one small adapter per factor, which is what the subtask asks for when it requires the two modules to be learned independently of each other.

The form of that adapter is what then decides how two of them can be combined. A LoRA adapter is not an extra piece of network but a matrix of the same shape as the weight it adapts, so adding it to that weight leaves the architecture untouched. Two adapters can therefore be summed and folded into the backbone together (Zhang et al., 2023), and merging in this sense meets all three demands the task makes. It needs no data and no coefficient to tune, so it applies in the zero-shot setting where there is nothing to fit. It gives back an ordinary model rather than an arrangement of several, so the composition is scored like any other checkpoint. And because the result is an ordinary model, another adapter can be trained on top of it, which is what the light supervised setting needs. How merging is applied and how the resulting models are scored is in §5, and the training settings are in §4.

Two alternatives serve as points of comparison, each dropping one property of this setup: steering vectors (Turner et al., 2023), an adapter obtained without any training, and logit-space composition (Liu et al., 2021), which combines output distributions without merging. Neither is part of the main results.

3 Subtask 2: test domain selection

Which genre. The probe of §2 already answers this. Every one of the six models finds Books harder than DGT, and by a margin far wider than anything that separates the models from each other. The left panel of Figure 2 shows it for EuroLLM-1.7B: the two curves barely touch anywhere. Literary text is where the backbones are weakest, and a

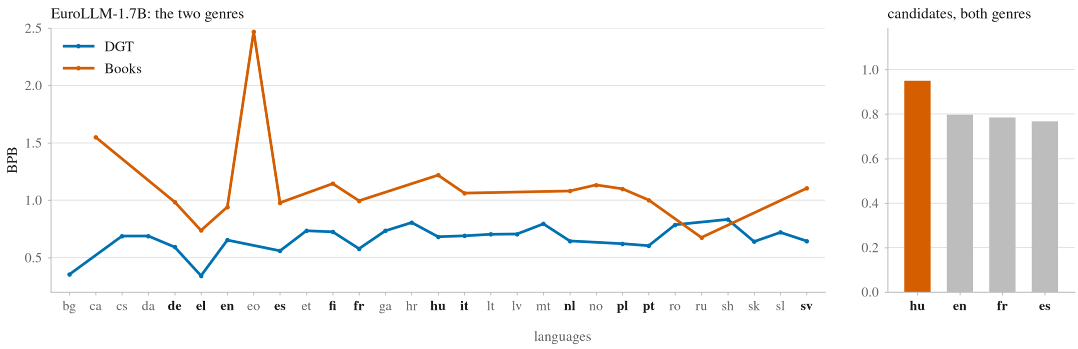

Figure 2: Left: EuroLLM-1.7B on every language of both datasets, showing how far apart the two genres sit. Right: the four candidate languages, each averaged over both genres. The other five models give the same picture.

study about adapting to a scarce domain is most informative where adaptation has something to do, so the target genre is Books and the language module will have to come from DGT.

Which language. The budget decides the shortlist before quality does. The subtask caps the training portion at 1M words, and whatever is left after that million goes to the test set. We wanted at least half a million words of test text, enough books that per-book comparisons mean something, which puts the bar for a candidate language at 1.5M words in Books. Sorting the Books languages by size shows how quickly that bites: only four clear it, en with 4.9M words, fr with 3.1M, hu with 2.7M and es with 2.1M. Those four are the shortlist.

Among them the choice is Hungarian, and the probe makes it unanimously. Averaged over both genres, all six models rank Hungarian hardest of the four, and by a clear margin over the other three, which sit close together. The right panel of Figure 2 shows this for EuroLLM-1.7B; the other five put the four in the same order. Hungarian is also the one candidate that is not Indo-European, it is agglutinative, and both backbones cut it into markedly more tokens per word than the other three. Those are three independent reasons pointing the same way, and they matter because a factorised method is meant for the cell where the backbone is least at home. We therefore take $l =$ Hungarian and $g =$ literary, the Hungarian portion of Books.

Split. The 28 Hungarian books are shuffled with a fixed seed and filled greedily into train up to 800k words, then dev up to 200k, with everything left over going to test. The unit is the whole book. That gives 10 books and 796,930 words for training, 2 books and 197,510 for dev, and 16 books and 1.70M words for test. Train and dev together stay inside the 1M cap, because a validation set used to pick checkpoints is information the model was built from and counting it separately would quietly raise the budget.

4 Subtask 3: independent module training

What each module sees. The subtask fixes both corpora. The genre module is trained on the target dataset without the target language, which is Books minus Hungarian: 15 languages, 130 documents, 13.8M words. The language module is trained on the target language of the other dataset, the Hungarian portion of DGT, which holds 42,283 documents and 84.6M words. Neither run ever sees the target domain, and neither sees anything the other one saw.

The two pools differ by a factor of six, and a module that has read six times more text is not a fair partner for one that has not. We therefore cap both at the same word budget, the smaller of the two, and sample the language corpus down to it. Within that budget each module keeps nine tenths for training and one tenth for choosing checkpoints, which comes to roughly 12.5M words of training text and 1.3M held out. Sampling is by whole documents with a fixed seed, and a language contributing only one document to Books keeps it in the training half, so no language appears solely in dev.

How they are trained. Every adapter in this report, including those trained on the target split, comes from one configuration: rank 16 on the seven projection matrices of every layer, $\alpha = 2r$, dropout 0.05, AdamW at a peak learning rate of

Configuration	EuroLLM-1.7B BPB	EuroLLM-1.7B SD	EuroLLM-1.7B Δ	Qwen3-4B BPB	Qwen3-4B SD	Qwen3-4B Δ
base	1.1684	–	–	1.1910	–	–
language adapter	1.1964	0.0001	$\uparrow$0.0280	1.2472	0.0006	$\uparrow$0.0562
genre adapter	1.1624	0.0001	$\downarrow$0.0060	1.1862	0.0002	$\downarrow$0.0048
direct adapter	1.1501	0.0008	$\downarrow$0.0183	1.1712	0.0004	$\downarrow$0.0198
composed adapter (zero shot)	1.1877	0.0002	$\uparrow$0.0193	1.2395	0.0006	$\uparrow$0.0485
composed adapter (few shot)	**1.1489**	0.0003	$\downarrow$0.0194	**1.1706**	0.0004	$\downarrow$0.0204

Table 2: BPB on the held-out Hungarian Books test split, 16 books, as the mean and standard deviation over three training seeds. Δ is against the backbone; $\downarrow$ improves, $\uparrow$ degrades. Bold and shading mark the best adapted model per backbone.

10^{-4} with 3% linear warmup and linear decay to zero, gradients clipped at norm 1.0, and an effective batch of 32 sequences of 1024 tokens. The factor modules see their corpus once; anything trained on the 1M-word target split sees it five times, because that split is an order of magnitude smaller. Every 25 updates the adapter is scored on its dev split and the best checkpoint so far is kept, so no run is early-stopped.

Training text is packed the usual way: the documents of a corpus are tokenized, joined into one stream with an EOS token between them, and cut into non-overlapping sequences of 1024 tokens. Nothing is prepended, so the first token of a sequence is context rather than something to predict.

Fixing one configuration is what makes the modules comparable later. Two adapters built with different α/r live on different scales, and summing them would weight one more heavily for a reason that has nothing to do with what they learned.

5 Subtask 4: composition and evaluation

5.1 Experiments

Baselines. One backbone and three adapters, trained as described in §4 and scored on the same 16 books: the base model, which says whether adapting helps at all; the language adapter alone and the genre adapter alone, which say what each factor is worth by itself; and the *direct adapter*, trained on the 1M-word target split, which says what is available if modularity is abandoned.

Against these we evaluate the composition in the two settings the subtask defines. *Zero-shot*: both modules merged into the backbone at full strength, nothing fitted. *Few-shot*: the merged model used as the starting point for training on the same 1M words. The few-shot model sees the same data, budget and hyperparameters as the direct adapter and differs only in where training starts, so the gap between the two isolates the value of composed initialisation.

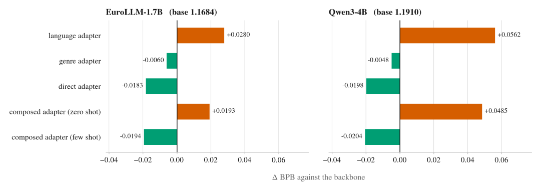

Figure 3: Every adapted model as a change against its own backbone, three-seed means. Green improves, vermilion degrades. The zero-shot merge inherits the language adapter's domain shift; both supervised models cross zero; the pattern repeats on two unrelated backbones.

Evaluation. Scoring follows the training setup in everything that affects the numbers: blocks of 1024 tokens, no overlap, nothing prepended, the first token of a block unscored. One deliberate difference: training joins documents into a stream, while scoring cuts each document on its own, so every block belongs to exactly one book and can be clustered by it. Every trained result is the mean of three seeds.

5.2 Results

Table 2 holds the main numbers, and Figure 3 draws them as changes against each backbone.

Baselines. The genre adapter helps on both backbones (-0.0060 and -0.0048 BPB): literary text in fifteen other languages transfers to Hungarian literature. The language adapter hurts on both ($+0.0280$ and $+0.0562$). This is the corpus problem of §4 arriving in the results: what the module learned about Hungarian is outweighed by what it learned about legal prose, and the harm is larger on the backbone that knew Hungarian less well to begin with. The direct adapter beats the base comfortably on both backbones and sets the mark the compositions have to meet.

Zero-shot. The merged model does not beat the base: addition is faithful, and one of the things being added is harmful. The composition itself still behaves. Against the language adapter alone, merging in the genre module recovers -0.0087 and -0.0077 BPB on the two backbones: the genre module's contribution survives being summed with a harmful partner, at nearly the same effect size on two unrelated models. The operator preserves what the modules carry; what it was given to carry is the problem.

Few-shot. Both supervised models clear the backbone by a wide margin and land within a hair of each other, the few-shot model ahead by -0.0011 and -0.0007 BPB. The direction is the same across seeds, but at this size the comparison is simply too close to call. What holds without qualification is that composed initialisation costs nothing, a start that was worse than the backbone zero-shot is no worse after adaptation.

5.3 Exploration

The main line fixes two choices: the module is a LoRA adapter, and the composer merges both updates into the weights. We varied each once, holding everything else fixed. Table 3 scores the three zero-shot compositions.

Changing the composer costs a little. Combining the two experts in logit space is worse on both backbones, and it gives up what merging provides for free: the result is no longer a single model, so the few-shot setting cannot be built on it. Changing the adapter costs everything. A steering vector is a fixed per-layer offset extracted without training, and adding two of them moves the hidden states so far off distribution that the model stops modelling Hungarian.

	EuroLLM-1.7B		Qwen3-4B	
Composition	BPB	Δ	BPB	Δ
LoRA + merge	1.1877	↑0.0193	1.2395	↑0.0485
LoRA + logit	1.1953	↑0.0269	1.2572	↑0.0662
steering + merge	4.3138	↑3.1454	7.9474	↑6.7563

Table 3: Zero-shot composition built three ways. Only the adapter type or the composer changes; corpora, budgets and protocol are those of §4. The shaded row is the one used in the main results.

6 Extension: a second target language

We repeated the study with $l' =$ French, one seed, both backbones, same budgets, splits, protocol and code path. French Books holds 29 books; after the same 1M-word cap the test set is 22 books and 2.1M words. Table 4 has the numbers.

The Hungarian conclusions replicate. The genre adapter helps (-0.0103 and -0.0105), and the language adapter hurts even for high-resource French ($+0.0130$, $+0.0200$), so the domain shift is a property of the recipe, not of Hungarian.

Two things change. First, the few-shot comparison that stayed open on Hungarian resolves: merged-then-adapted beats the direct adapter on EuroLLM by -0.0055 BPB and points the same way on Qwen. Second, the value of modularity inverts with backbone competence. On Hungarian the direct adapter dominated everything zero-shot. On French, where the backbone is already strong, it barely helps, while the genre adapter alone, cross-lingual transfer with no French literary text at all, is the best adapted model on both backbones. Where in-domain data is the bottleneck, supervision wins; where the backbone already covers the language, the modular factor is the larger source of gain.

Figure 4 puts the two languages side by side. The ordering is the same in both: the language adapter is worst, the genre adapter helps, the zero-shot merge sits between them, and the few-shot model is best or close to it. What changes is the scale. Everything that depends on the backbone knowing the language less well shrinks toward zero, most visibly the direct adapter, worth -0.0183 on Hungarian and nothing on French. The genre adapter is the one bar that moves the other way, and the only one never trained on a word of the target language.

On comparing the two domains: absolute BPB does not support it, since byte entropy differs across languages, so every claim above is a within-language contrast and the domains are compared at the level of conclusions.

7 Limitations

Books is a parallel corpus, so most of the genre corpus consists of translations of works that also exist in the target language; part of the genre adapter's contribution may be familiarity with the same works rather than with the genre. The merge

Configuration	EuroLLM-1.7B BPB	EuroLLM-1.7B Δ	Qwen3-4B BPB	Qwen3-4B Δ
base	0.9684	–	0.9001	–
language adapter	0.9814	↑0.0130	0.9200	↑0.0200
genre adapter	**0.9581**	↓**0.0103**	**0.8896**	↓**0.0105**
direct adapter	0.9694	↑0.0009	0.8983	↓0.0018
composed adapter (zero shot)	0.9703	↑0.0018	0.9075	↑0.0075
composed adapter (few shot)	0.9639	↓0.0045	0.8951	↓0.0050

Table 4: The study repeated on French Books, 22 held-out novels, one seed; budgets, splits and protocol unchanged. Δ is against the backbone.

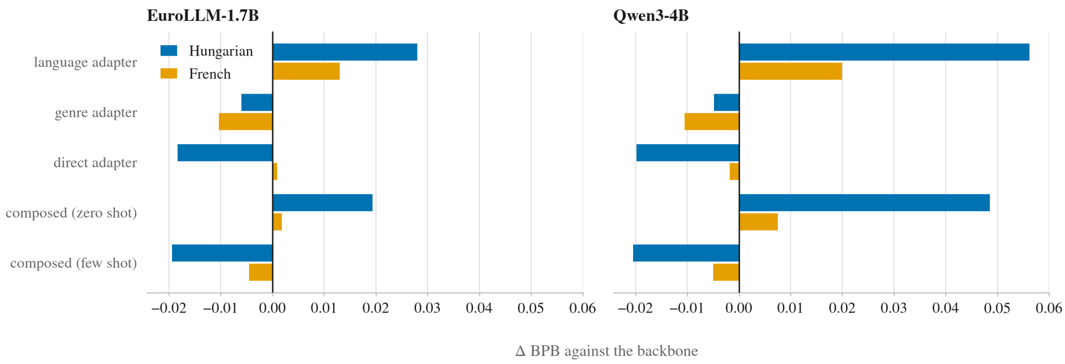

Figure 4: The five adapted models on both target languages, each as a change against its own backbone. Hungarian is the mean of three seeds, French one seed. Every effect shrinks toward zero on French except the genre adapter's, which grows.

adds both updates at equal weight, and we never checked whether an unequal weighting does better; tuning one would need held-out data that the zero-shot setting does not assume. Both backbones are small, and nothing here has been checked at a larger scale. All conclusions are language-modelling metrics.

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